

Class 1: Introduction, Causality, Self-selection Bias, Regression

Ricardo A Pasquini

April 2011

- Angrist's what should be the FAQs of a researcher: The basics of a good research.
 - While most econometrics courses focus on the details of empirical research, and take the choice of the topic as given.
 - But a coherent and doable research agenda is the basis.
- FAQ1: *Which is the causal effect of interest? (examples: schooling on wages, democratic institutions on growth)*

- FAQ2: *The ideal experiment that would be useful to study the causal question of interest.*

- FAQ2: *The ideal experiment that would be useful to study the causal question of interest.*
 - You cannot randomize education.. but..you can give incentives to potential dropouts to finish school. Angrist and Lavy (2007)

- FAQ2: *The ideal experiment that would be useful to study the causal question of interest.*
 - You cannot randomize education.. but..you can give incentives to potential dropouts to finish school. Angrist and Lavy (2007)
 - Also there are Fundamentally Unidentified Questions: Race or gender. No chromosomes can be changed.

- FAQ2: *The ideal experiment that would be useful to study the causal question of interest.*
 - You cannot randomize education.. but..you can give incentives to potential dropouts to finish school. Angrist and Lavy (2007)
 - Also there are Fundamentally Unidentified Questions: Race or gender. No chromosomes can be changed.
 - Or you can be very imaginative:

- FAQ2: *The ideal experiment that would be useful to study the causal question of interest.*
 - You cannot randomize education.. but..you can give incentives to potential dropouts to finish school. Angrist and Lavy (2007)
 - Also there are Fundamentally Unidentified Questions: Race or gender. No chromosomes can be changed.
 - Or you can be very imaginative:
 - Discrimination: What would have been the effect is that individual where black etc. Treating someone different because they believe they are different.

- FAQ2: *The ideal experiment that would be useful to study the causal question of interest.*
 - You cannot randomize education.. but..you can give incentives to potential dropouts to finish school. Angrist and Lavy (2007)
 - Also there are Fundamentally Unidentified Questions: Race or gender. No chromosomes can be changed.
 - Or you can be very imaginative:
 - Discrimination: What would have been the effect is that individual where black etc. Treating someone different because they believe they are different.
 - Bertrand and Mullainathan (2004) compared employers' responses to resumes with blacker-sounding and whiter-sounding ... names, like Lakisha and Emily (Fryer and Levitt, 2004, note that names may carry information about socioeconomic status as well as race.)

- FAQ2: *The ideal experiment that would be useful to study the causal question of interest.*
 - You cannot randomize education.. but..you can give incentives to potential dropouts to finish school. Angrist and Lavy (2007)
 - Also there are Fundamentally Unidentified Questions: Race or gender. No chromosomes can be changed.
 - Or you can be very imaginative:
 - Discrimination: What would have been the effect is that individual where black etc. Treating someone different because they believe they are different.
 - Bertrand and Mullainathan (2004) compared employers' responses to resumes with blacker-sounding and whiter-sounding ... names, like Lakisha and Emily (Fryer and Levitt, 2004, note that names may carry information about socioeconomic status as well as race.)
 - And there are other example with no apparent solution: Do children do better in school by virtue of having started school a little older?

- FAQ2: *The ideal experiment that would be useful to study the causal question of interest.*
 - You cannot randomize education.. but..you can give incentives to potential dropouts to finish school. Angrist and Lavy (2007)
 - Also there are Fundamentally Unidentified Questions: Race or gender. No chromosomes can be changed.
 - Or you can be very imaginative:
 - Discrimination: What would have been the effect is that individual where black etc. Treating someone different because they believe they are different.
 - Bertrand and Mullainathan (2004) compared employers' responses to resumes with blacker-sounding and whiter-sounding ... names, like Lakisha and Emily (Fryer and Levitt, 2004, note that names may carry information about socioeconomic status as well as race.)
 - And there are other example with no apparent solution: Do children do better in school by virtue of having started school a little older?
 - If they start older they are also... older... pure maturing effect.

- FAQ3: *What is your identification strategy?*
 - Identification Strategy: the manner in which a researcher uses observational data (i.e., data not generated by a randomized trial) to approximate a real experiment.

- FAQ3: *What is your identification strategy?*
 - Identification Strategy: the manner in which a researcher uses observational data (i.e., data not generated by a randomized trial) to approximate a real experiment.
 - Angrist and Krueger (1991) use a natural experiment to estimate the effects of finishing high school on wages.

- FAQ3: *What is your identification strategy?*
 - Identification Strategy: the manner in which a researcher uses observational data (i.e., data not generated by a randomized trial) to approximate a real experiment.
 - Angrist and Krueger (1991) use a natural experiment to estimate the effects of finishing high school on wages.
 - Compulsory laws ask students to remain in school until 16 or 17th birthday.

- FAQ3: *What is your identification strategy?*
 - Identification Strategy: the manner in which a researcher uses observational data (i.e., data not generated by a randomized trial) to approximate a real experiment.
 - Angrist and Krueger (1991) use a natural experiment to estimate the effects of finishing high school on wages.
 - Compulsory laws ask students to remain in school until 16 or 17th birthday.
 - Individuals born in the beginning of the year start school at an older age, so they can drop out after completing less schooling than individuals born near the end of the year.

- FAQ4: *What is your mode of statistical inference?*
 - The answer to this question describes the population to be studied, the sample to be used, and the assumptions made when constructing standard errors.

The experimental Ideal

- Consider causal “if-then” question, and an example: Do hospitals make people healthier?

The experimental Ideal

- Consider causal “if-then” question, and an example: Do hospitals make people healthier?
 - Assume we are studying poor elderly population who use for primary care.

The experimental Ideal

- Consider causal “if-then” question, and an example: Do hospitals make people healthier?
 - Assume we are studying poor elderly population who use for primary care.
 - Empirical approach: compare the health of those who have attended hospital and those who have not.

The experimental Ideal

- Consider causal “if-then” question, and an example: Do hospitals make people healthier?
 - Assume we are studying poor elderly population who use for primary care.
 - Empirical approach: compare the health of those who have attended hospital and those who have not.
 - Using National Health Interview Survey (NHIS), includes a question “During the past 12 months, was the respondent a patient in a hospital overnight?” which we can use to identify recent hospital visitors. The NHIS also asks “Would you say your health in general is excellent (1), very good (2), good, fair, poor (5) ?”

The experimental Ideal

Group	Sample Size	Mean Health Status	Std. Error
Hospital	7774	2.79	0.014
Non-Hospital	90049	2.07	0.003

- The difference in the means is 0.71, a large and highly significant contrast in favor of the non-hospitalized, with a t-statistic of 58.9.
- It should follow that hospitals make people sicker.
- Hospitals are crowded with sick people, infections, dangerous machines...
- Still, it's easy to see why this comparison should not be taken at face value: people who go to the hospital are probably less healthy to begin with.

The experimental Ideal

- Lets define: Treatment $D_i = \{0, 1\}$ and for a given individual i the potential outcome.

$$\text{potential outcome} \begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases}$$

The experimental Ideal

- Lets define: Treatment $D_i = \{0, 1\}$ and for a given individual i the potential outcome.

$$\text{potential outcome} \begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases}$$

- Where $Y_{1i} - Y_{0i}$ is the causal effect of interest. However, in the practice we will not be able to see both Y_{1i} and Y_{0i} for a given individual.

The experimental Ideal

- Lets define: Treatment $D_i = \{0, 1\}$ and for a given individual i the potential outcome.

$$\text{potential outcome} \begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases}$$

- Where $Y_{1i} - Y_{0i}$ is the causal effect of interest. However, in the practice we will not be able to see both Y_{1i} and Y_{0i} for a given individual.
- Note that the observed outcome, Y_i , can be written in terms of potential outcomes as

$$\begin{aligned} Y_i &= \begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases} \\ &= Y_{0i} - (Y_{0i} - Y_{1i})D_i \end{aligned}$$

The experimental Ideal I

- Once defined the random variable, we can use it to learn something about the expectation. The comparison of average health conditional on hospitalization status is formally linked to the average causal effect by the equation below:

$$E[Y_i | D_i = 1] - E[Y_i | D_i = 0]$$

Observed difference in average health

The experimental Ideal

- Adding and subtracting $E[Y_{0i} \mid D_i = 1]$ (a theoretically well defined term)

$$= \underbrace{E[Y_{1i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 1]}_{\text{Average Effect of the Treatment on the Treated}} + \underbrace{E[Y_{0i} \mid D_i = 0] - E[Y_{0i} \mid D_i = 1]}_{\text{Selection Bias}}$$

The experimental Ideal

- Adding and subtracting $E[Y_{0i} | D_i = 1]$ (a theoretically well defined term)

$$= \underbrace{E[Y_{1i} | D_i = 1] - E[Y_{0i} | D_i = 1]}_{\text{Average Effect of the Treatment on the Treated}} + \underbrace{E[Y_{0i} | D_i = 0] - E[Y_{0i} | D_i = 1]}_{\text{Selection Bias}}$$

- Notice that the *Selection Bias* measures the differences between the groups that exist *in the absence of the treatment*.

The experimental Ideal

- Adding and subtracting $E[Y_{0i} | D_i = 1]$ (a theoretically well defined term)

$$= \underbrace{E[Y_{1i} | D_i = 1] - E[Y_{0i} | D_i = 1]}_{\text{Average Effect of the Treatment on the Treated}} + \underbrace{E[Y_{0i} | D_i = 0] - E[Y_{0i} | D_i = 1]}_{\text{Selection Bias}}$$

- Notice that the *Selection Bias* measures the differences between the groups that exist *in the absence of the treatment*.
- Notice: because the sick are more likely than the healthy to seek treatment, those who were hospitalized have worse Y_{0i} 's, making selection bias negative in this example.

The experimental Ideal

- Adding and subtracting $E[Y_{0i} | D_i = 1]$ (a theoretically well defined term)

$$= \underbrace{E[Y_{1i} | D_i = 1] - E[Y_{0i} | D_i = 1]}_{\text{Average Effect of the Treatment on the Treated}} + \underbrace{E[Y_{0i} | D_i = 0] - E[Y_{0i} | D_i = 1]}_{\text{Selection Bias}}$$

- Notice that the *Selection Bias* measures the differences between the groups that exist *in the absence of the treatment*.
- Notice: because the sick are more likely than the healthy to seek treatment, those who were hospitalized have worse Y_{0i} 's, making selection bias negative in this example.
- The first term $E[Y_{1i} | D_i = 1] - E[Y_{0i} | D_i = 1]$ can be written as: $E[Y_{1i} - Y_{0i} | D_i = 1]$, and interpreted as the average effect on the hospitalized given that they have been hospitalized.

Random Assignment I

- The **random assignment** solves the selection problem: guarantees the independence between D_i and Y_{0i} , allowing to cancel the second term and yields:

$$E[Y_{1i} | D_i = 1] - E[Y_{0i} | D_i = 1] = E[Y_{1i} - Y_{0i}]$$

- Examples of selection bias in research:
 - *Evaluation of government-subsidized training programs.*
 - These are programs that provide a combination of classroom instruction and on-the-job training for groups of disadvantaged workers such as the long-term unemployed, drug addicts, and ex-offenders. The idea is to increase employment and earnings. Paradoxically, studies based on non-experimental comparisons of participants and non-participants often show that after training, the trainees earn less than plausible comparison groups (see, e.g., Ashenfelter, 1978; Ashenfelter and Card, 1985; Lalonde 1995).

- Here too, selection bias is a natural concern since subsidized training programs are meant to serve men and women with low earnings potential. Not surprisingly, therefore, simple comparisons of program participants with non-participants often show lower earnings for the participants. In contrast, evidence from randomized evaluations of training programs generate mostly positive effects (see, e.g., Lalonde, 1986; Orr, et al, 1996).

Random Assignment

- *Effect of class size*

Random Assignment

- *Effect of class size*
- Observational studies often reflect the fact that small sizes are grouped with disadvantaged skills, so selection bias is a problem to evaluate the effect.

Random Assignment

- *Effect of class size*
- Observational studies often reflect the fact that small sizes are grouped with disadvantaged skills, so selection bias is a problem to evaluate the effect.
- The Tennessee STAR Program is an example of randomized evaluation.

Random Assignment

- *Effect of class size*
- Observational studies often reflect the fact that small sizes are grouped with disadvantaged skills, so selection bias is a problem to evaluate the effect.
- The Tennessee STAR Program is an example of randomized evaluation.
 - It cost about \$12 million and was implemented for a cohort of kindergartners in 1985/86.

Random Assignment

- *Effect of class size*
- Observational studies often reflect the fact that small sizes are grouped with disadvantaged skills, so selection bias is a problem to evaluate the effect.
- The Tennessee STAR Program is an example of randomized evaluation.
 - It cost about \$12 million and was implemented for a cohort of kindergartners in 1985/86.
 - The average class size in regular Tennessee classes in 1985/86 was about 22.3.

Random Assignment

- *Effect of class size*
- Observational studies often reflect the fact that small sizes are grouped with disadvantaged skills, so selection bias is a problem to evaluate the effect.
- The Tennessee STAR Program is an example of randomized evaluation.
 - It cost about \$12 million and was implemented for a cohort of kindergartners in 1985/86.
 - The average class size in regular Tennessee classes in 1985/86 was about 22.3.
 - The experiment assigned students to one of three treatments: small classes with 13-17 children, regular classes of 22-25 children and a part-time teacher's aide, or regular classes with a full time teacher's aide.

Random Assignment

- The first question to ask about a randomized experiment, or any experiment like this in general, is whether the randomization successfully balanced subject's characteristics across the different treatment groups. To assess this, it's common to compare pre-treatment outcomes or other covariates across groups. Not in this case:

Random Assignment

Table 2.2.1: Comparison of treatment and control characteristics in the Tennessee STAR experiment

Students who entered STAR in kindergarten				
Variable	Small	Regular	Regular/Aide	Joint <i>P</i> -value
1. Free lunch	.47	.48	.50	.09
2. White/Asian	.68	.67	.66	.26
3. Age in 1985	5.44	5.43	5.42	.32
4. Attrition rate	.49	.52	.53	.02
5. Class size in kindergarten	15.10	22.40	22.80	.00
6. Percentile score in kindergarten	54.70	48.90	50.00	.00

- The *P*-value in the last column is for the *F*-test of equality of variable means across all three groups.

Random Assignment

Table 2.2.1: Comparison of treatment and control characteristics in the Tennessee STAR experiment

Students who entered STAR in kindergarten				
Variable	Small	Regular	Regular/Aide	Joint <i>P</i> -value
1. Free lunch	.47	.48	.50	.09
2. White/Asian	.68	.67	.66	.26
3. Age in 1985	5.44	5.43	5.42	.32
4. Attrition rate	.49	.52	.53	.02
5. Class size in kindergarten	15.10	22.40	22.80	.00
6. Percentile score in kindergarten	54.70	48.90	50.00	.00

- The *P*-value in the last column is for the *F*-test of equality of variable means across all three groups.
- Class sizes are significantly lower in the assigned-to-be-small class rooms, which means that the experiment succeeded in creating the desired variation.

Random Assignment

- A typical problem in implementation: If many of the parents of children assigned to regular classes had effectively lobbied teachers and principals to get their children assigned to small classes, the gap in class size across groups would be much smaller.

Random Assignment

- A typical problem in implementation: If many of the parents of children assigned to regular classes had effectively lobbied teachers and principals to get their children assigned to small classes, the gap in class size across groups would be much smaller.
- In practice, the difference in means between treatment and control groups can be obtained from a regression of test scores on dummies for each treatment group. The estimated treatment-control differences for kindergartners, reported in Table 2.2.2 (derived from Krueger, 1999, Table 5), show a small-class effect of about 5 to 6 percentile points.

Random Assignment

Table 2.2.2: Experimental estimates of the effect of class-size assignment on test scores

Explanatory variable	(1)	(2)	(3)	(4)
Small class	4.82	5.37	5.36	5.37
	(2.19)	(1.26)	(1.21)	(1.19)
Regular/aide class	.12	.29	.53	.31
	(2.23)	(1.13)	(1.09)	(1.07)
White/Asian (1 = yes)	–	–	8.35	8.44
			(1.35)	(1.36)
Girl (1 = yes)	–	–	4.48	4.39
			(.63)	(.63)
Free lunch (1 = yes)	–	–	-13.15	-13.07
			(.77)	(.77)
White teacher	–	–	–	-.57
				(2.10)
Teacher experience	–	–	–	.26
				(.10)
Master's degree	–	–	–	-0.51
				(1.06)
School fixed effects	No	Yes	Yes	Yes
R ²	.01	.25	.31	.31

- The effect size is about 2σ where σ is the standard deviation of the percentile score in kindergarten.

Random Assignment

Table 2.2.2: Experimental estimates of the effect of class-size assignment on test scores

Explanatory variable	(1)	(2)	(3)	(4)
Small class	4.82 (2.19)	5.37 (1.26)	5.36 (1.21)	5.37 (1.19)
Regular/aide class	.12 (2.23)	.29 (1.13)	.53 (1.09)	.31 (1.07)
White/Asian (1 = yes)	–	–	8.35 (1.35)	8.44 (1.36)
Girl (1 = yes)	–	–	4.48 (.63)	4.39 (.63)
Free lunch (1 = yes)	–	–	-13.15 (.77)	-13.07 (.77)
White teacher	–	–	–	-.57 (2.10)
Teacher experience	–	–	–	.26 (.10)
Master's degree	–	–	–	-0.51 (1.06)
School fixed effects	No	Yes	Yes	Yes
R ²	.01	.25	.31	.31

- The effect size is about 2σ where σ is the standard deviation of the percentile score in kindergarten.
- The small-class effect is significantly different from zero, while the regular/aide effect is small and insignificant.

- The STAR study also highlights the logistical difficulty, long duration, and potentially high cost of randomized trials.

- The STAR study also highlights the logistical difficulty, long duration, and potentially high cost of randomized trials.
- We hope to find natural or quasi-experiments that mimic a randomized trial by changing the variable of interest while other factors are kept balanced. Can we always find a convincing natural experiment? Of course not.

- The STAR study also highlights the logistical difficulty, long duration, and potentially high cost of randomized trials.
- We hope to find natural or quasi-experiments that mimic a randomized trial by changing the variable of interest while other factors are kept balanced. Can we always find a convincing natural experiment? Of course not.
- Angrist and Lavy (1999) relies on the fact that in Israel, class size is capped at 40. Therefore, a child in a fifth grade cohort of 40 students ends up in a class of 40 while a child in fifth grade cohort of 41 students ends up in a class only half as large because the cohort is split. Since students in cohorts of size 40 and 41 are likely to be similar on other dimensions such as ability and family background, we can think of the difference between 40 and 41 students enrolled as being “as good as randomly assigned.”

Regression Analysis of Experiments

- Suppose that the treatment effect is the same for everyone:

$$Y_{1i} - Y_{0i} = \rho$$

Regression Analysis of Experiments

- Suppose that the treatment effect is the same for everyone:

$$Y_{1i} - Y_{0i} = \rho$$

- Recall the equation for the observed outcome:

$$Y_i = Y_{0i} - (Y_{0i} - Y_{1i})D_i$$

Regression Analysis of Experiments

- Suppose that the treatment effect is the same for everyone:

$$Y_{1i} - Y_{0i} = \rho$$

- Recall the equation for the observed outcome:

$$Y_i = Y_{0i} - (Y_{0i} - Y_{1i})D_i$$

- We will the derive a specification for a regression model and derive what does the estimation yields.

Regression Analysis of Experiments

- Suppose that the treatment effect is the same for everyone:
 $Y_{1i} - Y_{0i} = \rho$
- Recall the equation for the observed outcome:
 $Y_i = Y_{0i} - (Y_{0i} - Y_{1i})D_i$
- We will then derive a specification for a regression model and derive what the estimation yields.
- Adding and subtracting the constant $E[Y_{0i}]$ (the expected value of the outcome in the absence of the treatment - the average for the population that has not been treated):

$$Y_i = E[Y_{0i}] + Y_{0i} - (Y_{0i} - Y_{1i})D_i - E[Y_{0i}]$$

Regression Analysis of Experiments

- Suppose that the treatment effect is the same for everyone:

$$Y_{1i} - Y_{0i} = \rho$$

- Recall the equation for the observed outcome:

$$Y_i = Y_{0i} - (Y_{0i} - Y_{1i})D_i$$

- We will then derive a specification for a regression model and derive what the estimation yields.
- Adding and subtracting the constant $E[Y_{0i}]$ (the expected value of the outcome in the absence of the treatment - the average for the population that has not been treated):

$$Y_i = E[Y_{0i}] + Y_{0i} - (Y_{0i} - Y_{1i})D_i - E[Y_{0i}]$$

- We can rearrange it to obtain a one variable linear equation model:

$$Y_i = \underbrace{E[Y_{0i}]}_{\text{constant}=\alpha} + \underbrace{(Y_{1i} - Y_{0i})}_{=\rho \text{ by assumption on constant effects}} D_i + \underbrace{Y_{0i} - E[Y_{0i}]}_{=\eta_i \text{ random term}}$$

Regression Analysis of Experiments

- The regression model is:

$$Y_i = \alpha + \rho D_i + \eta_i$$

Regression Analysis of Experiments

- The regression model is:

$$Y_i = \alpha + \rho D_i + \eta_i$$

- Note that η_i is the random part of Y_{0i}

Regression Analysis of Experiments

- The regression model is:

$$Y_i = \alpha + \rho D_i + \eta_i$$

- Note that η_i is the random part of Y_{0i}
- As outcome of the regression we obtain $\hat{\alpha}$ and $\hat{\rho}$

Regression Analysis of Experiments

- The regression model is:

$$Y_i = \alpha + \rho D_i + \eta_i$$

- Note that η_i is the random part of Y_{0i}
- As outcome of the regression we obtain $\hat{\alpha}$ and $\hat{\rho}$
- Evaluating the conditional expectations of the model, we can infer that the estimated coefficient yield the desired effect plus the selection bias. Evaluating when the treatment is on and off yields:

$$\begin{aligned} E[Y_i \mid D_i = 1] &= \alpha + \rho + E[\eta_{0i} \mid D_i = 1] \\ E[Y_i \mid D_i = 0] &= \alpha + E[\eta_{0i} \mid D_i = 0] \end{aligned}$$

Regression Analysis of Experiments

- The regression model is:

$$Y_i = \alpha + \rho D_i + \eta_i$$

- Note that η_i is the random part of Y_{0i}
- As outcome of the regression we obtain $\hat{\alpha}$ and $\hat{\rho}$
- Evaluating the conditional expectations of the model, we can infer that the estimated coefficient yield the desired effect plus the selection bias. Evaluating when the treatment is on and off yields:

$$E[Y_i | D_i = 1] = \alpha + \rho + E[\eta_{0i} | D_i = 1]$$

$$E[Y_i | D_i = 0] = \alpha + E[\eta_{0i} | D_i = 0]$$

- Subtracting both expressions we obtain the expression for the difference of the expected outcomes between the treatment and control groups:

$$E[Y_i | D_i = 1] - E[Y_i | D_i = 0] = \underbrace{\rho}_{\text{Treatment Effect}} + E[\eta_{0i} | D_i = 1] - E[\eta_{0i} | D_i = 0] \quad \underbrace{\hspace{10em}}_{\text{Selection Bias}}$$

Regression Analysis of Experiments

- The expression above tell us that the difference of expected outcomes between treatment and control groups will yield the treatment effect plus a bias term. The bias term will depend on the correlation between the treatment variable D_i and the error of the regression. Given that:

$$E[\eta_{0i} \mid D_i = 1] - E[\eta_{0i} \mid D_i = 0] = E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]$$

Regression Analysis of Experiments

- The expression above tell us that the difference of expected outcomes between treatment and control groups will yield the treatment effect plus a bias term. The bias term will depend on the correlation between the treatment variable D_i and the error of the regression. Given that:

$$E[\eta_{0i} \mid D_i = 1] - E[\eta_{0i} \mid D_i = 0] = E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]$$

- Once again, if the expected potential outcome of not receiving the outcome in the treatment group is different from the expected potential outcome of not receiving the outcome in the control group, we will obtained a bias estimator of the treatment effect.

Regression Analysis of Experiments

- The expression above tell us that the difference of expected outcomes between treatment and control groups will yield the treatment effect plus a bias term. The bias term will depend on the correlation between the treatment variable D_i and the error of the regression. Given that:

$$E[\eta_{0i} \mid D_i = 1] - E[\eta_{0i} \mid D_i = 0] = E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]$$

- Once again, if the expected potential outcome of not receiving the outcome in the treatment group is different from the expected potential outcome of not receiving the outcome in the control group, we will obtained a bias estimator of the treatment effect.
- Recalling the previous selection biases examples, in the hospital allegory, those who were treated had poorer health outcomes in the no-treatment state, while in the Angrist and Lavy (1999) study, students in smaller classes tend to have intrinsically lower test scores.

References I

- Angrist, J., (1990), "Lifetime Earnings and the Vietnam Era Draft Lottery: Evidence from Social Security Administrative Records," American Economic Review, 80, 313-335.
- Angrist, J., G. Imbens and D. Rubin (1996), "Identification of Causal Effects Using Instrumental Variables," Journal of the American Statistical Association, 91, 444-472.
- Bertrand and Mullainathan (2004)
- Fryer and Levitt (2004)
- Imbens, G., and J. Angrist (1994), "Identification and Estimation of Local Average Treatment Effects," Econometrica, Vol. 61, No. 2, 467-476.
- Lalonde, R.J., (1986), "Evaluating the Econometric Evaluations of Training Programs with Experimental Data," American Economic Review, 76, 604-62

- Manski, C., (1996), “Learning about Treatment Effects from Experiments with Random Assignment of Treatments,” The Journal of Human Resources, 31(4): 709-73.
- Rubin, Donald B. (1973): “Matching to Remove Bias in Observational Studies,” Biometrics, 29, 159–83. (1974): “Estimating the Causal Effects of Treatments in Randomized and Non-Randomized Studies,” Journal of Educational Psychology, 66, 688–701.
- (1977): “Assignment to a Treatment Group on the Basis of a Covariate,” Journal of Educational Statistics, 2, 1–26.
- (1991): “Practical Implications of Modes of Statistical Inference for Causal Effects and the Critical Role of the Assignment Mechanism,” Biometrics, 47, 1213–34.